

Statistics for Data Analytics

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Continuous Assessment 2

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Contents

1. Introduction.....	3
2. Logistic Regression Analysis of Customer Churn.....	3
2.1 Overview of Telco Customer Churn Dataset.....	3
2.2 Selected Input and Output Variables for Telco Churn Model	3
2.3 Model Training and Significant variables.	4
2.4 Prediction Using the Trained Model	4
2.6 Model Evaluation Using Confusion Matrix and Predicted Values	5
2.7 Confusion Matrix and Model Evaluation Results.....	5
2.8 Summary.....	5
3. Bayesian Inference for Poisson-Distributed Data	6
3.1 Likelihood Function.....	6
3.2 Conjugate Prior for λ	6
3.3 Posterior Distribution of λ	6
3.4 Minimum Bayes Risk Estimator.....	7
3.5 Summary.....	8
4. Time Series Analysis of the NASDAQ Composite Index.....	9
4.1 Data Description.....	9
4.2 Stationarity Assessment.....	9
4.3 Identification of AR and MA Orders	10
4.4 ARIMA Model Estimation and Selection	12
4.5 Forecasting.....	13
4.6 Residual Diagnostics	14
4.7 Summary.....	14
5. Conclusion	15
References	16

1. Introduction

Task 1 required a Generalised Linear Model to be fitted to a real-world customer level dataset. Task 2 required a Bayesian analysis of data assumed to follow a Poisson distribution, including derivation of the likelihood function, selection of a conjugate prior, derivation of the posterior distribution and calculation of the minimum-risk Bayes estimator. Task 3 required a time series analysis of a stock market dataset to fit an appropriate ARIMA model and to produce a ten-step-ahead forecast.

2. Logistic Regression Analysis of Customer Churn

2.1 Overview of Telco Customer Churn Dataset

Telco churn dataset was used for task one (*Telco Customer Churn*). It contained both continuous and categorical variables. The original dataset contained 7043 rows. However, after converting the Total charges variable into numeric, some missing values were identified. These incomplete records were removed using the `na.omit()` function. 11 rows were removed leaving a cleaned dataset with 7032 complete observations for analysis. The output variable was customer churn, represented by `y`. Customer who churned were coded as 1, while customers who did not churn were coded as 0. This was a binary classification problem, where the aim was to predict whether a customer is likely to leave the company or stay. The appropriate GLM was a binary logistic regression model with a binomial family and logit link function because the response variable had two outcomes (McCullagh and Nelder, 1989).

2.2 Selected Input and Output Variables for Telco Churn Model

The table 1 below shows the output variable and the input variables used for the binary logistic regression analysis. These variables included both continuous and categorical predictors, which made dataset suitable for the GLM model.

Code Variable	Original Variable	Type	Description
y	Churn	Output (binary)	Whether the customer churned (1) or not (0)
tenure	Tenure	Continuous	Number of months the customer has stayed with the company
monthly_charges	MonthlyCharges	Continuous	Amount charged to the customer each month
total_charges	TotalCharges	Continuous	Total amount charged to the customer over the subscription period
contract	Contract	Categorical	Contract type: month-to-month, one year or two year
internet_service	InternetService	Categorical	Type of internet service subscribed to
payment_method	PaymentMethod	Categorical	Method of payment, e.g. electronic check, mailed check or bank transfer

Code Variable	Original Variable	Type	Description
online_security	OnlineSecurity	Categorical	Whether the customer has an online security add-on

Table 1: Input and output variables selected for the churn model.

2.3 Model Training and Significant variables.

The data was divided into two parts: 80% was used as the training dataset, while the remaining 20% was used for testing the data. A random seed was set using `set.seed(32)` to make the results reproducible. The model was trained using the training dataset with the binomial family.

The model summary was used to examine the statistical significance of the individual model coefficients. The hypothesis test was carried out at the 5% significance level. The null hypothesis states that an input variable has no significant effect on customer churn. The alternative hypothesis states that an input variable has significant effect on the customer churn. Variables with p-values less than 0.05 were considered statistically significant. The estimated model coefficients were also obtained using `coef(fit)`, showing the direction and size of the effect of each predictor on customer churn. The coefficient estimates showed the direction of the relationship between each predictor and churn. A positive coefficient increased the log-odds of churn, while a negative coefficient reduced the log-odds of churn, holding other variables constant.

At the 5% significance level, the model coefficients were examined to assess their contribution to explaining customer churn. For categorical predictors, the p-values were interpreted at the individual category level because the logistic regression output provides separate coefficients for each category relative to a reference category. The reduced model selected using `stepAIC()` did not retain Monthly Charges, suggesting that it contributed less to the model than the predictors that remained.

2.4 Prediction Using the Trained Model

The trained model was used to predict churn outcomes for the test dataset. The model first produced predicted probabilities, and a threshold of 0.5 was applied. Predicted probabilities greater than 0.5 were classified as churn, while probabilities of 0.5 or below were classified as non-churn.

2.5 Reduced Model Selection

The `stepAIC()` method was used to produce a simpler logistic regression model by removing variables that added less value to the analysis. Monthly Charges was not included in the reduced model, while Tenure, Total Charges, Contract, Internet Service, Payment Method and Online Security were retained. This reduced model was then used for prediction and evaluation because it provided a simpler model for predicting customer churn.

2.6 Model Evaluation Using Confusion Matrix and Predicted Values

The predicted values were compared with the actual values using a confusion matrix. Accuracy was calculated to measure the proportion of correct predictions. Recall and precision were also calculated to provide more detail about how the model identified customers who churned.

2.7 Confusion Matrix and Model Evaluation Results

The performance of the AIC-selected reduced logistic regression model was evaluated using a confusion matrix, accuracy, recall and precision. The confusion matrix compares the actual churn values with the predicted churn values, while the evaluation metrics show how well the model classified customers in the test dataset. The results in table 2 were based on the single 80% training and 20% testing split generated using `set.seed(32)`, and the model performance may vary slightly if a different random split is used.

Result	Meaning	Value
True negative	Actual 0 correctly predicted as 0	922
False positive	Actual 0 incorrectly predicted as 1	97
False negative	Actual 1 incorrectly predicted as 0	180
True positive	Actual 1 correctly predicted as 1	208
Accuracy	$(922 + 208) / (922 + 97 + 180 + 208)$	80.31%
Recall	$208 / (208 + 180)$	53.61%
Precision	$208 / (208 + 97)$	68.20%

Table 2: Confusion matrix and evaluation metrics for the reduced model on the test set.

2.8 Summary.

The model correctly classified 80.31% of the customers in the test set, which shows that it performed well overall. However, the recall value of 53.61% means that it identified only a little over half of the customers who actually left the company, so a considerable number of churners were missed. The precision of 68.20% also shows that around two-thirds of the customers predicted to churn had actually done so. Overall, the model gives a useful prediction of customer churn, but it could be improved further by adjusting the classification threshold or using methods that focus more on identifying customers who are likely to leave.

3. Bayesian Inference for Poisson-Distributed Data

This looked at a sample $x_1, \dots, x_{10}$, assumed to be independently and identically distributed according to a $\text{Poisson}(\lambda)$ distribution. The task required the likelihood function to be derived, an appropriate conjugate prior to be chosen for λ , the posterior distribution to be found, and the minimum-risk Bayes estimator of λ to be computed (Gelman *et al.*, 2013).

3.1 Likelihood Function

For a single Poisson observation, the probability mass function was $f(x_i | \lambda) = \lambda^{x_i} e^{-\lambda} / x_i!$. Since the ten observations were independent, the joint likelihood is the product of the individual mass functions:

$$L(\lambda | x_1, \dots, x_{10}) = \prod_{i=1}^{10} [\lambda^{x_i} e^{-\lambda} / x_i!] = \lambda^{\sum x_i} e^{-10\lambda} / \prod x_i!$$

Because the term $\prod x_i!$ does not depend on λ , it was to be treated as a constant of proportionality when the likelihood is combined with a prior, giving:

$$L(\lambda | x_1, \dots, x_{10}) \propto \lambda^{\sum x_i} e^{-10\lambda}$$

3.2 Conjugate Prior for λ

A conjugate prior is a prior distribution which, when combined with the likelihood, yields a posterior distribution of the same family. For a Poisson likelihood, the Gamma distribution was the conjugate prior for the rate parameter λ . The hyperparameters were chosen as $\alpha = 2$ and $\beta = 1$, giving a prior of $\lambda \sim \text{Gamma}(2, 1)$, with density $f(\lambda) \propto \lambda^{\alpha-1} e^{-\beta\lambda} = \lambda e^{-\lambda}$. The Gamma (2,1) distribution was chosen as an illustrative prior for λ . It has a mean of 2 and a variance of 2, allowing some uncertainty in the possible value of λ before the observed data are taken into account.

3.3 Posterior Distribution of λ

The posterior was proportional to the product of the likelihood and the prior:

$$f(\lambda | x) \propto L(\lambda | x) \times f(\lambda) \propto \lambda^{\sum x_i} e^{-10\lambda} \times \lambda e^{-\lambda} = \lambda^{\sum x_i + 1} e^{-11\lambda}$$

This was the kernel of a Gamma distribution with shape parameter $\sum x_i + 2$ and rate parameter 11, so that:

$$\lambda | x_1, \dots, x_{10} \sim \text{Gamma}(\sum x_i + 2, 11)$$

To obtain numerical results, a sample of size 10 was simulated in R from a Poisson (4) distribution using `set.seed(123)`, giving the observed values 3, 6, 3, 6, 7, 1, 4, 7, 4, 4, with $\sum x_i = 45$. Substituting this into the expression above gives a posterior of $\lambda | x \sim \text{Gamma}(47, 11)$. The maximum likelihood estimate of λ was found by checking a range of possible values between 0.01 and 10. The likelihood was highest when $\lambda=4.50$. This result matched the usual Poisson estimate, which was the sample mean. Since the ten observations added up to 45, the sample mean was $45/10=4.5$.

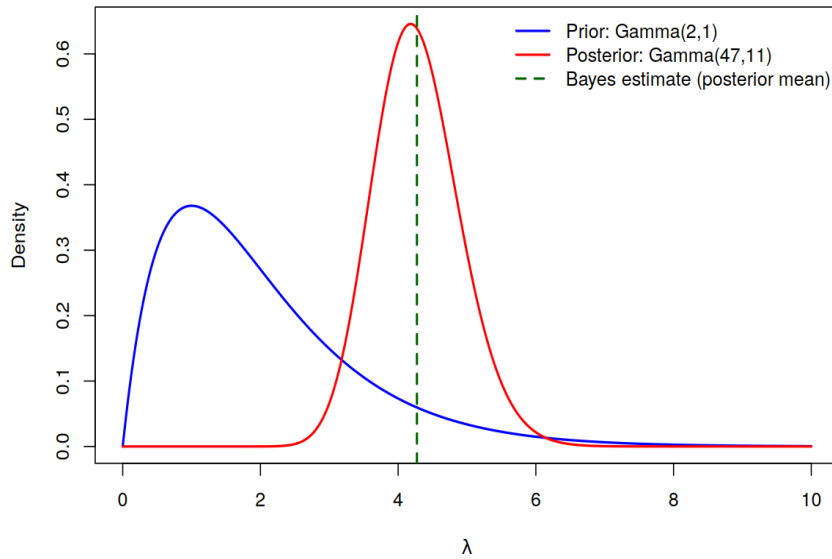


Figure 1: Prior distribution, Gamma (2,1), and posterior distribution, Gamma (47,11), of λ , with the Bayes estimate (posterior mean) marked.

Figure 1 shows the prior and posterior densities together. The posterior was both narrower and shifted to the right of the prior, reflecting the fact that the observed data, with a sample mean of 4.5, have updated the initial belief about λ towards higher values while substantially reducing the uncertainty around it.

3.4 Minimum Bayes Risk Estimator

The minimum Bayes risk estimator depends on the loss function used. Under squared-error loss, the Bayes estimator is the mean of the posterior distribution; under absolute-error loss, it is the posterior median. For a Gamma (α^* , β^*) posterior, the mean is α^*/β^* . Using $\alpha^* = 47$ and $\beta^* = 11$:

Quantity	Formula	Value
Bayes estimate (squared-error loss)	$E(\lambda x) = \alpha^*/\beta^*$	4.2727
Bayes estimate (absolute-error loss)	Median of Gamma (47,11)	4.2425
Posterior variance	α^*/β^{*2}	0.3884
95% credible interval	2.5th–97.5th percentiles of Gamma (47,11)	[3.139, 5.578]

Table 4: Bayes estimates and posterior summary statistics for λ .

The posterior mean of 4.2727 as shown in table 4 was adopted as the minimum Bayes risk estimator of λ under squared-error loss, this being the most commonly used loss function in practice. It is close to, but slightly lower than, the maximum likelihood estimate of 4.5, illustrating how the Gamma (2,1) prior pulls the Bayes estimate towards the prior mean of 2

relative to a purely data-driven estimate. The 95% credible interval for λ was between 3.139 and 5.578. Compared with the prior distribution, the posterior was more concentrated because it combined the original prior information with the evidence from the ten observed values. This reduced the uncertainty about the likely value of λ .

3.5 Summary

The likelihood function for an iid Poisson sample was derived, a Gamma (2,1) conjugate prior was specified for λ , and the resulting posterior was shown analytically to be Gamma ($\sum x_i + 2$, $n + 1$). Applying this to a simulated sample of size 10 gave a posterior of Gamma (47, 11), from which the minimum Bayes risk estimator under squared-error loss was calculated as 4.2727, with a 95% credible interval of [3.139, 5.578].

4. Time Series Analysis of the NASDAQ Composite Index

Task 3 required a time series analysis of a stock market dataset, covering stationarity testing, identification of candidate AR and MA orders from the autocorrelation structure, selection of an ARIMA model, and a ten-step-ahead forecast.

4.1 Data Description

The dataset used for this analysis consists of the daily closing prices of the NASDAQ Composite Index obtained from Yahoo Finance using the quantmod package in R (*NASDAQ Composite (^IXIC) Charts, Data & News*). Quantmod (Quantitative Financial Modelling) provides direct access to financial market data, making it suitable for time series analysis and forecasting. The data used is the NASDAQ composite index, which is an index listed on the NASDAQ stock exchange. The index tracks the performance of the companies listed on the NASDAQ stock exchange (*What Is NASDAQ? - Wealthspire*). Most of the stocks on NASDAQ are technological companies and this is a good data set as it aggregates the performance of various companies' stocks. The NASDAQ index has a ticker of “^IXIC” and the variable used is the closing prices which are stored as “xts” because it represents daily observations (1,255 observations) between 1 January, 2021 to 31 December, 2025.

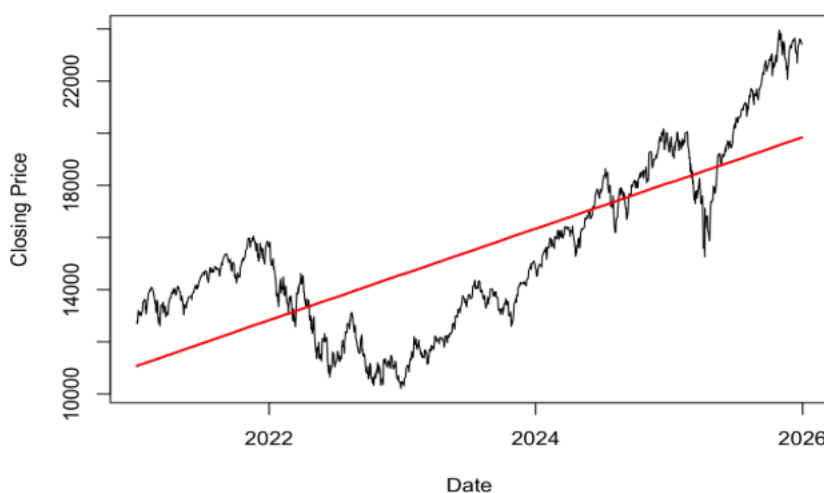


Figure 2: NASDAQ Composite closing price, 2021–2025, with a fitted linear trend line.

The original daily closing prices were plotted to visualise the data and how it fits the simple linear regression and overlays the fitted trend line. Figure 2 shows an overall upward trend in NASDAQ index in the period under review. The fitted regression line confirms a positive trend, and this shows that the average closing price increased within the 5 years and this confirms non-stationarity due to the presence of a trend. Therefore, differencing is required before an ARIMA model can be fitted.

4.2 Stationarity Assessment

An ARIMA model requires a stationary series, so the NASDAQ closing price was first assessed for stationarity in both mean and variance.

Stationarity in Mean: Figure 2 shows a clear upward trend over the study period between the closing price and the date. The fitted regression line also has a positive slope, indicating that the average closing price increases over time. This shows non-stationary in the mean.

Stationarity in Variance: Figure 2 suggests slightly greater variability in the later part of the series. Although this increase is not sufficiently pronounced to conclude non-stationarity in the variance by visual inspection alone, a logarithmic transformation was applied as a precaution to stabilise the variance before modelling.

Because the series was judged non-stationary in the mean, first differencing was applied, using `diff()` followed by `na.omit()` to remove the resulting leading missing value.

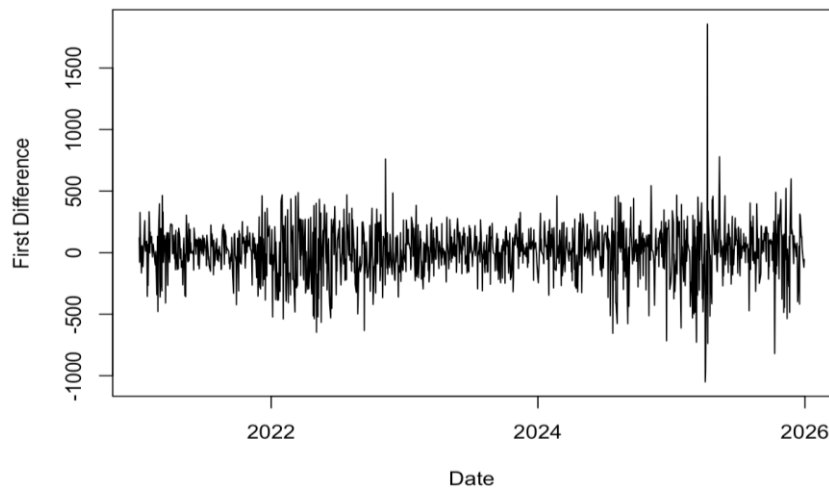


Figure 3: First-differenced NASDAQ closing price series.

Figure 3 shows that the first-differenced series moves around zero and no longer has the clear upward trend seen in the original data. The Augmented Dickey–Fuller test gave a test statistic of -10.741 and a p-value below 0.01 . Because this p-value was lower than the 5% significance level, the null hypothesis of a unit root was rejected. This suggests that the series became stationary after one differencing step, so an order of differencing of $d=1$ was used in the ARIMA models.

4.3 Identification of AR and MA Orders

The autocorrelation function (ACF) and partial autocorrelation function (PACF) of the raw series were examined first, primarily to illustrate its non-stationarity rather than to identify AR or MA orders directly, since ACF and PACF are only meaningfully interpreted once a series is stationary.

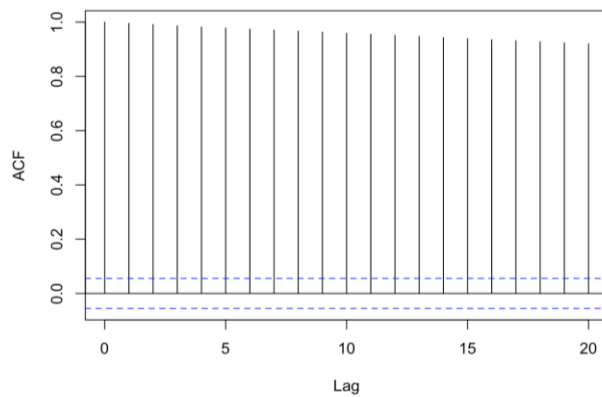


Figure 4a: ACF of the raw NASDAQ closing price series.

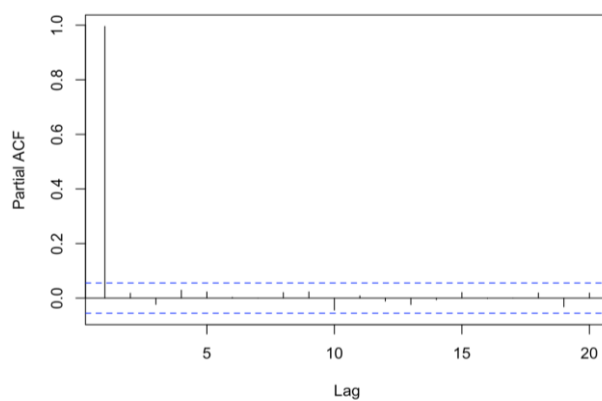


Figure 4b: PACF of the raw NASDAQ closing price series.

As Figure 4a shows, the ACF of the raw series remains close to 1 and decays only very slowly across the 20 lags plotted, which is typical of a non-stationary, highly persistent series. The corresponding PACF in Figure 4b shows a single large spike at lag 1, with all subsequent lags falling within the confidence bounds, indicating that each day's closing price is very strongly related to the previous day's price, as would be expected of this kind of series.

The ACF and PACF of the differenced series were then examined, and it is this pair of plots that was used to identify candidate AR and MA orders, since the differenced series had already been shown to be stationary.

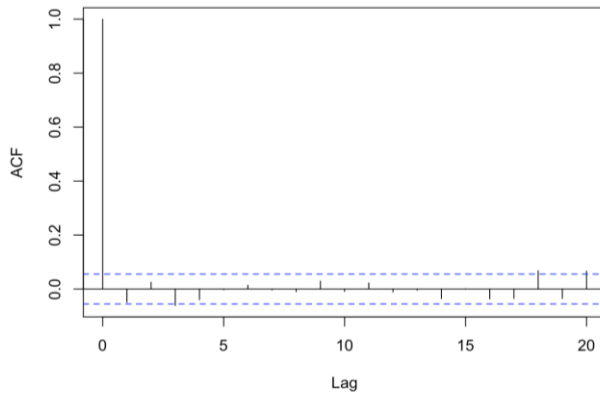


Figure 5a: ACF of the first-differenced NASDAQ series.

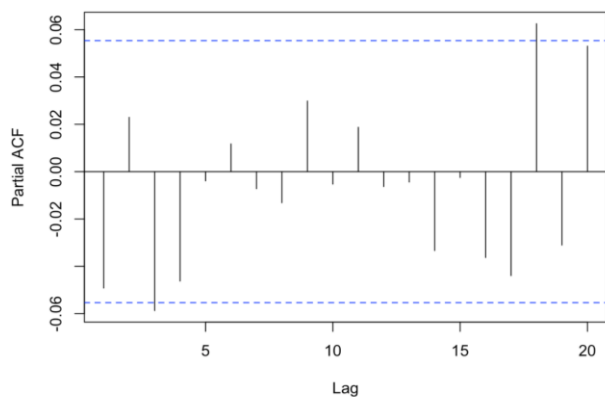


Figure 5b: PACF of the first-differenced NASDAQ series.

Following differencing, nearly all of the autocorrelations in Figure 5a and the partial autocorrelations in Figure 5b fall within the 95% confidence bounds, with only isolated, marginal spikes at a few scattered lags. This pattern, close to that of white noise, suggests that little systematic AR or MA structure remains once the trend has been removed, and is broadly consistent with the differenced series behaving like a near random walk. This motivated a small set of parsimonious candidate models: an AR(1) specification, ARIMA(1,1,0); an MA(1) specification, ARIMA(0,1,1); and a mixed ARMA(1,1) specification, ARIMA(1,1,1), each fitted on top of a single order of differencing.

4.4 ARIMA Model Estimation and Selection

The daily NASDAQ closing prices do not show an obvious repeating seasonal pattern, therefore it is treated as non-seasonal and a non-seasonal ARIMA model was fitted.

The three candidate models, ARIMA(1,1,0), ARIMA(0,1,1) and ARIMA(1,1,1), were each fitted to the NASDAQ closing price series using the `Arima()` function, with the differencing handled internally by specifying $d = 1$ in the model order. The models were compared using the Akaike Information Criterion (AIC), which balances model fit against model complexity, with a lower AIC indicating a preferred model.

As an independent check on this manual identification, the `auto.arima()` function was also used to search automatically over a range of ARIMA specifications. Consistent with the near white-

noise appearance of the differenced ACF and PACF, `auto.arima()` selected an $ARIMA(0,1,1)$ model with drift as the best-fitting specification, corresponding to the $MA(1)$ candidate identified manually from the differenced PACF/ACF, with an additional drift term to capture the small average upward movement in the series over the sample period. This agreement between the manually identified candidate models and the automatically selected model provides some reassurance that the model order was correctly identified from the ACF/PACF diagnostics.

Although $ARIMA(1,1,0)$ produced the lowest AIC among the manually specified models, the automatically selected $ARIMA(0,1,1)$ with drift achieved a slightly lower overall AIC (16975.39) and was therefore selected as the final forecasting model.

4.5 Forecasting

Both the manually specified $ARIMA(1,1,0)$ model and the automatically selected $ARIMA(0,1,1)$ with drift model were used to generate a ten-step-ahead forecast of the NASDAQ closing price, using the `forecast()` function with $h = 10$. The `forecast()` plotting method automatically overlays the forecast, together with its 80% and 95% prediction intervals, on the historical series, allowing the projected values to be assessed directly against recent history.

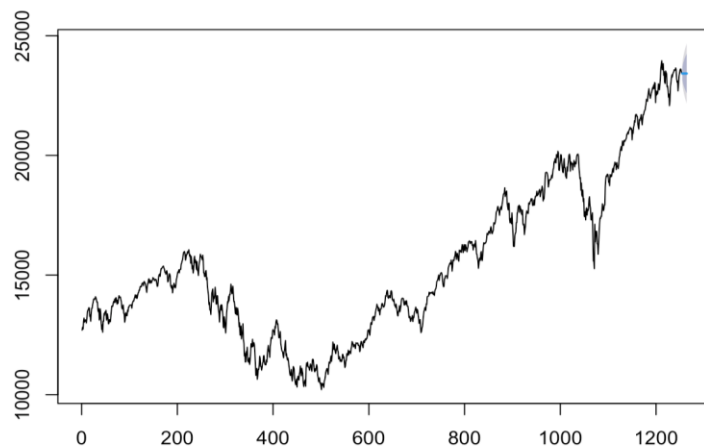


Figure 6a: Ten-step-ahead forecast from the manually specified $ARIMA(1,1,0)$ model.

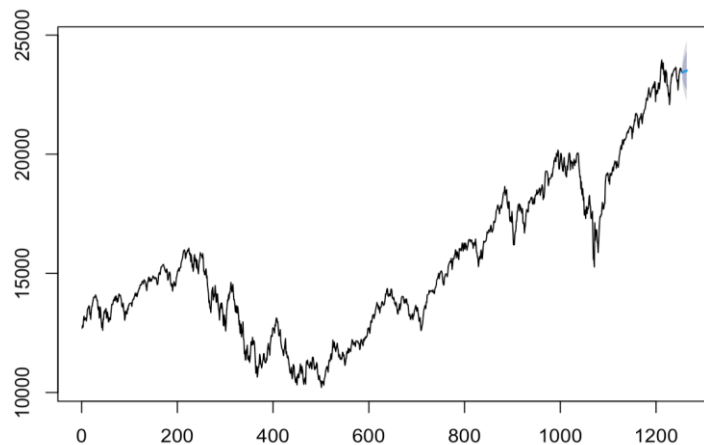


Figure 6b: Ten-step-ahead forecast from the automatically selected ARIMA (0,1,1) with drift model.

Both models as shown in Figure 6a and 6b produce a similar, near-flat short-horizon forecast that continues from the last observed value of the series, with a prediction interval that widens steadily over the ten-step forecast horizon, reflecting the compounding uncertainty inherent in forecasting a highly persistent, near random-walk financial series. This behaviour is expected for a differenced series with little remaining autocorrelation structure: the best short-term forecast is close to a continuation of the most recent level, adjusted only slightly by the small drift term in the case of the ARIMA (0,1,1) with drift specification.

4.6 Residual Diagnostics

Residual diagnostics were carried out using the `checkresiduals()` function to assess whether the fitted models adequately captured the structure of the data (Cryer and Chan, 2009). The residual plots fluctuate randomly around zero with no obvious systematic pattern, while the residual ACF indicates that almost all autocorrelations lie within the 95% confidence limits.

The Ljung–Box test returned p-values of 0.4301 for the ARIMA(1,1,0) model and 0.4216 for the ARIMA(0,1,1) with drift model. Since both p-values exceed the 5% significance level, there is no evidence of remaining autocorrelation in the residuals. This suggests that both models adequately describe the data, although the ARIMA(0,1,1) with drift model was preferred because it achieved the lowest AIC.

4.7 Summary

The NASDAQ Composite closing price series was shown to be non-stationary in the mean, evidenced by its visible upward trend, a slowly decaying ACF and a large single-lag PACF spike, and this was confirmed formally by an Augmented Dickey-Fuller test on the first-differenced series (Dickey-Fuller statistic -10.741 , $p < 0.01$). The near white-noise appearance of the differenced ACF and PACF motivated three parsimonious candidate ARIMA models, and automatic model selection using `auto.arima()` converged on an ARIMA(0,1,1) with drift specification, consistent with the manually identified MA(1) candidate. This model, along with the manually specified ARIMA(1,1,0) alternative, was used to produce a ten-step-ahead forecast with widening prediction intervals, reflecting the high persistence typical of daily financial closing prices.

The residual diagnostics confirmed that the residuals behaved like white noise, indicating that the selected ARIMA(0,1,1) with drift model provides an adequate representation of the NASDAQ closing price series and is suitable for short-term forecasting.

5. Conclusion

This report has presented three distinct statistical analyses carried out for Continuous Assessment 2. In Task 1, a binary logistic regression model was used to predict customer churn in a telecommunications dataset, achieving 80.31% accuracy on a held-out test set, with tenure, total charges, contract type, internet service, payment method and online security all identified as statistically significant predictors. In Task 2, Bayesian methods were applied to a Poisson-distributed sample, deriving the likelihood function analytically, specifying a $\text{Gamma}(2,1)$ conjugate prior, and obtaining a $\text{Gamma}(47,11)$ posterior distribution, from which a minimum Bayes risk estimate of $\lambda = 4.2727$ was calculated under squared-error loss. In Task 3, the Box-Jenkins methodology was applied to the NASDAQ Composite Index ('Box-Jenkins Methodology in R', 2025), confirming non-stationarity in the raw series, identifying an $\text{ARIMA}(0,1,1)$ with drift model as the best fit, and producing a ten-step-ahead forecast. Taken together, the three tasks demonstrate the application of generalised linear modelling, Bayesian inference and time series forecasting to three different, realistic data analytics problems.

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